

In this chapter, we deep dive into the macroeconomic trends, employment scenario, international benchmarking and sectoral trends for five sectors.

3.1 Manufacturing

The Indian manufacturing sector is undergoing significant transformation, driven by various macroeconomic factors, technological advancements and evolving global dynamics. This section provides a comprehensive analysis of sectoral trends, benchmarking against international practices, and insights drawn from both primary interviews and secondary research. The focus is on understanding the future of jobs in the manufacturing sector, highlighting emerging trends, challenges and the promising opportunities that lie ahead.

3.1.1 Macroeconomic trends

A Key statistics

India's manufacturing sector is on the brink of substantial growth, expected to generate an additional 6.33% in new jobs during the first half of FY 2024-25²⁶. Notably, the manufacturing, engineering, and infrastructure sectors are among the top three with the highest proportion of employers expanding their workforce, with 66% planning to increase headcount. This expansion is driven by government reforms like the Production Linked Incentive (PLI) scheme, which has already attracted significant investments in automobiles, electronics, and textiles. Additionally, increased capital investments and rise in

mergers and acquisitions, particularly in the automobile, electronics, and textiles categories, indicate strong future prospects for the sector.

As a key contributor to India's economy, the manufacturing sector accounts for



16-17%
of GDP as of 2021



market size projected to reach
US\$1 trillion by 2025



GVA (Gross Value Added)
at current prices was
US\$ 110.48 billion
in the first quarter of FY24²⁷



compound annual
growth rate (CAGR) of
3.32% expected
between 2024 and 2029²⁸

Between April 2000 and December 2023, the automobile sector attracted FDI inflows of US\$36.26 billion, while the chemical manufacturing sector (excluding fertilizers) received FDI inflows of US\$22.14 billion. These figures underscore the sector's critical role in India's economy and its potential for future expansion, with substantial FDI inflows reflecting global confidence in India's manufacturing capabilities²⁹ and reinforcing the sector's importance as a major job provider.

B Employment scenario

As of 2024, the manufacturing sector employs approximately 27.3 million people³⁰. This sector has consistently expanded its employment opportunities, driven by growing industrial activities and supportive government initiatives.

C Key drivers of growth and government schemes

The Union Budget 2024 has solidified manufacturing as a cornerstone of India's path to becoming a developed nation, unveiling a comprehensive package of incentives and policy reforms. These include significant investments in industrial infrastructure, the expansion of the National Industrial Corridor Development Programme, and strategic adjustments to customs duties to bolster domestic manufacturing and exports. These measures aim to enhance India's global competitiveness and support the 'Make in India' initiative, laying a strong foundation for continued growth³¹.

Key initiatives and developments include:



Production Linked Incentive (PLI)

Scheme: The Interim Union Budget 2024-2025 saw a significant boost in allocations for the PLI Scheme, with a 360% increase to INR6,903 crore (US\$830 million) for the semiconductors and display manufacturing ecosystem, and a 623% rise to INR3,500 crore (US\$421 million) for the automobile sector.



Electric Vehicles (EV) expansion: The government has placed a strong emphasis on promoting the transition to electric vehicles (EV) by expanding EV charging networks, creating opportunities for small vendors in manufacturing and maintenance.



Defence allocation: The Ministry of Defence's budget was increased to INR621,541 crore (US\$74.78 billion), highlighting a commitment to enhancing domestic manufacturing capabilities in the defence sector.



The National Manufacturing Policy (NMP) aims to increase the sector's GDP contribution to 25% and create 100 million new jobs. This policy includes establishing National Investment and Manufacturing Zones (NIMZs) and offering

incentives for green technologies. While there are challenges, such as bureaucratic hurdles and the need for continuous upskilling, these initiatives underscore the government's commitment to sustainable growth in manufacturing.



SANKALP³² (Skills Acquisition and Knowledge Awareness for Livelihood Promotion) program, which strengthens institutional mechanisms for skill development and enhances access to quality, market-relevant training. Similarly, the **SAMARTH**³³ (Scheme for Capacity Building in Textile Sector) scheme is focused on providing skill development training in the textiles sector, a significant component of the manufacturing industry, ensuring the workforce is equipped to meet evolving demands.



The Vishwakarma³⁴ Scheme is designed to enhance the skills of traditional artisans and craftsmen through modern training programs, integrating them into the contemporary manufacturing sector.

These measures are designed to enhance India's global competitiveness and support the 'Make in India' initiative, laying a strong foundation for the manufacturing sector's continued expansion.

3.1.2 International benchmark

India's adoption of Industry 4.0 technologies, while growing, remains behind global leaders like Germany, Japan and the USA.

India's position in the global manufacturing sector is strong, ranking among the top in the Global Manufacturing Risk Index (GMRI) due to its cost-effectiveness, workforce availability, and market potential. India's labor costs are notably lower than in several key Asian markets, and it ranks fifth in global manufacturing output with a 2.78% share.

Adopting international best practices, such as those outlined in the Smart Industry Readiness Index (SIRI) and the Global Lighthouse Network, can further enhance India's manufacturing capabilities. These frameworks emphasize digital transformation, sustainable practices, and the development of advanced skills, positioning India for greater competitiveness in the global market.



3.1.3 Sectoral trends

A Emerging technologies in the sector

The anticipated impact of AI in areas like quality control and customer service is driving a shift in perception, indicating future growth. Industry 4.0 technologies like IoT-enabled predictive maintenance and supply chain optimization are already enhancing efficiency by reducing downtime and improving order fulfilment.

- AI is set to revolutionize agile manufacturing and quality control, though challenges such as high implementation costs and the need for skilled labor remain.
- Digital Public Infrastructure (DPI)³⁷ plays a crucial role in supporting seamless operations, including secure data exchanges and real-time processing, further boosting manufacturing efficiency.

B Workforce trends

In the manufacturing sector, hiring trends highlight an increasing demand for technically skilled professionals with qualifications in engineering, computer science, and specialized fields such as robotics and automation. Employers are actively seeking candidates with experience in Industry 4.0 technologies, including IoT, AI and advanced analytics. Additionally, skills in lean manufacturing, quality control and supply chain management are highly valued. The focus is on building a workforce capable of adapting to rapidly evolving technological environments and driving innovation and efficiency improvements.

Workforce mobility is another key factor shaping the job landscape in manufacturing. Increased mobility facilitates the exchange of skills and best practices across regions, fostering innovation and efficiency.

3.1.4

Future of jobs: insights from industry leaders

The primary survey has reached out to industries with strength of more than 500 employees working with their presence in regional, national and international markets.

Emerging job roles in the sector

The future of jobs in the manufacturing sector is poised to be shaped by roles that capitalize on advanced technologies. According to the survey, 60% of respondents anticipate significant growth in jobs for AI and robotics technicians, as well as IoT specialists, 40% foresee high demand for sustainable manufacturing engineers and digital supply chain analysts.

The survey further reveals that 40% of respondents believe skills in robotic programming, advanced analytics, and Internet-of-Things connectivity will be most in demand for future manufacturing roles. Only 20% expect a high demand for AI and machine learning application skills, which presents an interesting contrast, highlighting the evolving landscape of skill requirements in the sector.

Impact of GenAI, AI, sector-specific technology, international mobility, DPI, climate and green jobs on the job and skilling ecosystem

The impact generated by generative AI and AI technologies in the manufacturing sector is likely to be very strong. The use of these technologies may likely result in creating new job roles and altering the nature of existing ones, particularly in the case of supply chain

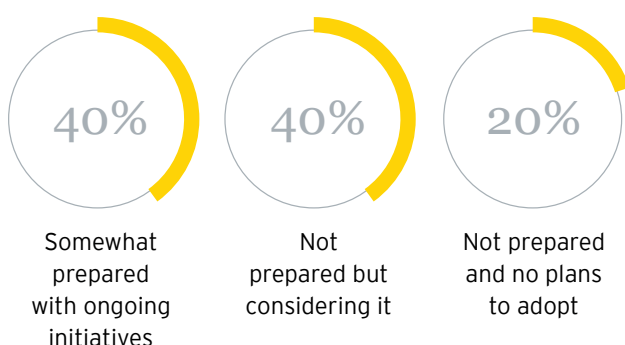
optimization and predictive maintenance. The survey revealed concerns related to sector preparedness in picking up these technologies. As much as 40% of the respondents registered ongoing initiatives in AI adoption preparedness, while 40% mentioned that organizations were not prepared at all and only thinking about it. By itself, this presents an enormous gap between recognition of AI potential and readiness towards its realization. Secondly, majority respondents highlighted the major challenges in AI integration are workforce's resistance, and lack of technical skills. The other factor identified was data privacy and security concerns. This is also related to secondary research, emphasizing a need for upskilling and reskilling initiative to face these challenges. The gap that ultimately exists between the projected level of influence that AI will have and the levels of preparedness and usage actually being at work within the manufacturing sector speaks to the fact that a more concerted approach needs to be taken with regards to the way in which such technologies will be integrated.

The surveyed organizations are increasingly adopting sustainability practices, with organizations implementing programs such as energy-efficient machinery (50%), waste reduction (33%), and sustainable sourcing (33%). AI and generative AI are expected to significantly impact supply chain optimization (44%), predictive maintenance (36%), and production line automation (27%).

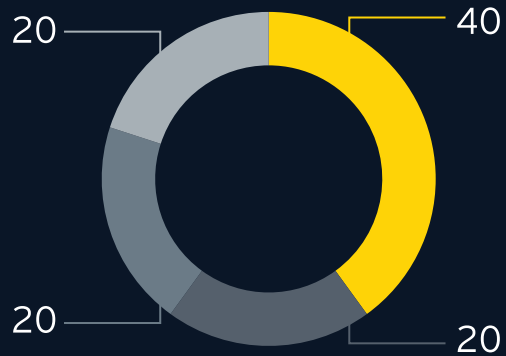
Suggested programmatic and policy interventions

Most of the industry representatives felt that the program and policy interventions had to match the dynamism of the sector. 80% of the respondents felt that governmental support in creating renewable energy infrastructure would be beneficial, 40% mentioned the importance of stringent environmental regulations to encourage sustainability. Equally, 40% believe that the grants for acquiring new equipment and software would be the most effective government incentives towards technological advancement. Out of the total workforce sample, 80% preferred a supportive government partnership with the private while none suggested the government to be at the frontline in training and education. This testifies to the preference for public-private collaboration over government initiatives in a situation where the urgency for skill development is well recognized.

Preparedness of AI adoption



Government incentives or subsidies can be/ have been most beneficial for technological advancements



- Grants for purchasing new equipment or software
- Tax credits for research and development
- Incentives for adopting sustainable practices
- None

Challenges manufacturing companies face in collaborating with educational and skilling institutions (respondents could select multiple options)

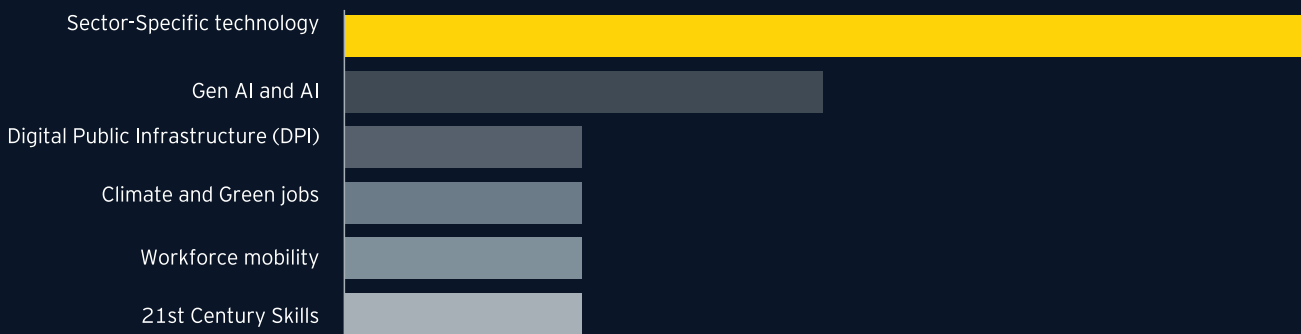
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Lack of awareness about schemes and programs

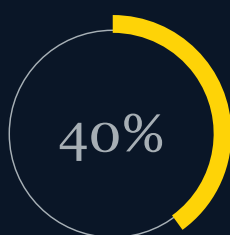
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Misalignment of training outcomes with industry needs

Areas skilling institutions needs to focus on:



What role should skilling institutions play in preparing the workforce for the adoption of new technologies?



Collaborative role in designing industry-specific courses



Primary role in providing foundational and advanced tech training



Supportive role with on-the-job training provided by employers



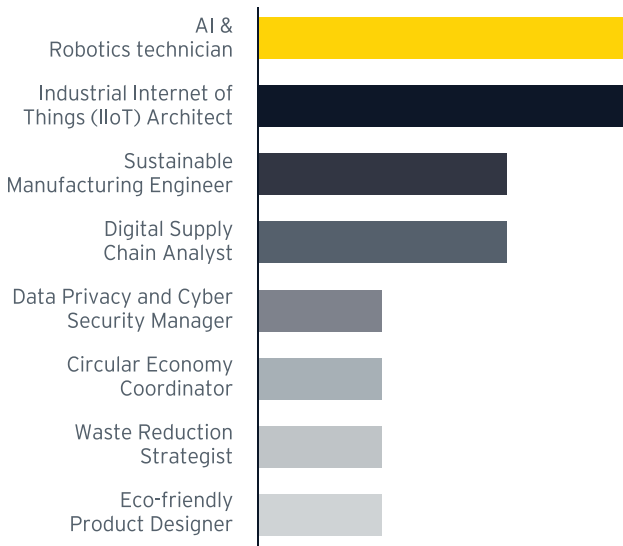
Advisory role in identifying future skill needs

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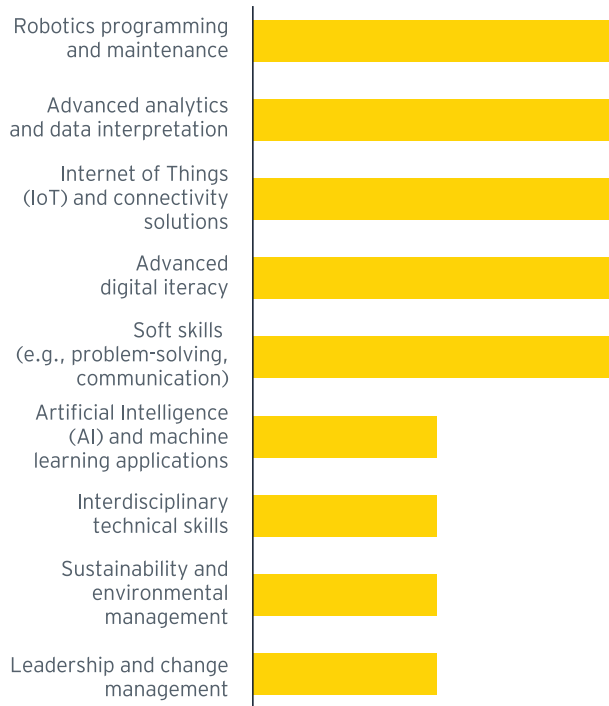
Overall impact on job roles and skills in the sector

The key challenges as identified in the primary research include the need for better alignment between the industry, skill development institutions, and government. 80% of the surveyed believed continuous learning and upskilling are very important. Majority of the respondents believed that lack of awareness about schemes and programs is the major challenge manufacturing companies face in collaborating with educational and skilling institutions. This points at the need for richer, stronger connections in training initiatives between industry and academia.

Emerging job roles in manufacturing sector



Skill sets that will be most in demand for future manufacturing job roles:



The future of jobs in the manufacturing sector in India is being influenced by a confluence of economic growth, technological advances and sustainability imperatives. While the sector is on a promising path toward growth and innovation, challenges such as skill gaps, technological readiness, and workforce resistance have to be accommodated. The findings of this survey call for a more coordinated and focused approach toward workforce development that aligns industry needs with skill development initiatives and government support.

